



Chapter 34-Port Security



Introduction to Port Security

👉 **Port Security** is a security feature of Cisco switches that helps control **which devices are allowed to connect** to a switch port.



Simple Definition:

- It controls **which MAC addresses can enter a port**
- If an **unauthorized MAC address** tries to connect → action is taken 🚨



Key Features of Port Security

- ✓ Controls **source MAC addresses**
- ✓ Limits number of devices per port
- ✓ Can **block unauthorized access**
- ✓ Protects network from attacks



Default Behavior

- By default:
 - Only **1 MAC address is allowed**
 - If unknown MAC comes:
 - Port goes into **err-disabled state** ❌
-

Real-Life Example

Think of a **security guard at a building** :

- Only **authorized people (MAC addresses)** can enter
- If unknown person tries:
 -  Entry blocked
 -  Alarm triggered

 Port Security = **Security guard of switch port**

Why Use Port Security?

Control Access

- Only trusted devices can connect

Protect Against Attacks

Example: DHCP Starvation Attack

- Attacker sends **thousands of fake MAC addresses**
- DHCP server gives IPs → pool exhausted 

Another Problem:

- MAC table overflow

 Solution:

✓ Limit number of MAC addresses per port

Important Note

 MAC Address Spoofing is possible:

- Tools like **T-MAC Changer** can change MAC address
 - So Port Security is helpful but **not 100% secure**
-

Enabling Port Security

Problem:

Command rejected: GigabitEthernet0/1 is a dynamic port.

Solution:

Port must be:

✓ Access OR Trunk (NOT dynamic)

Steps:

```
interface g0/1
switchport mode access
switchport port-security
```

Verify Mode

```
show interface g0/1 switchport
```

Verification Commands

```
show port-security interface g0/1
show interfaces status
```

Re-enabling Disabled Port

Manual Recovery

```
interface g0/1
shutdown
no shutdown
```

◆ Automatic Recovery

show errdisable recovery

errdisable recovery cause psecure-violation

errdisable recovery interval 180

👉 Default recovery: **every 5 minutes**

✓ After 3 minutes → port comes back automatically



Violation Modes (VERY IMPORTANT)

There are **3 modes** 👉

● 1. Shutdown (Default)

- Port goes **err-disabled**
- Traffic dropped ❌
- Logs generated 📄
- Manual/auto recovery needed

✓ Violation counter = 1

● 2. Restrict

- Traffic from unknown MAC = ❌ dropped
 - Port stays **UP**
 - Logs generated 📄
 - Counter increases 📊
-

● 3. Protect

- Traffic dropped ❌
 - Port stays **UP**
 - ❌ No logs
 - ❌ No counter increase
-

🧠 Easy Comparison

Mode	Port Status	Logs	Counter
Shutdown	Down ❌	Yes	1
Restrict	Up ✅	Yes	Increment
Protect	Up ✅	No	No

🔧 Configuration Examples

◆ Restrict Mode

```
switchport port-security
switchport port-security mac-address 000a.000a.000a
switchport port-security violation restrict
```

◆ Protect Mode

```
switchport port-security
switchport port-security mac-address 000a.000a.000a
switchport port-security violation protect
```

🔍 Verify

```
show port-security interface g0/1
```



Secure MAC Address Aging

◆ Default Behavior

- Aging time = **0 minutes**
 - MAC addresses **never expire**
-

◆ Set Aging Time

switchport port-security aging time 30

◆ Aging Types

● Absolute

- Timer starts after learning MAC
- Removed after time ends (even if active)

● Inactivity

- Timer resets on activity
- Removed only if idle

switchport port-security aging type inactivity

◆ Static MAC Aging

switchport port-security aging static



Real-Life Example

- Absolute = **Parking ticket time** 🕒
- Inactivity = **WiFi auto logout after inactivity** 📶

Sticky MAC Address

◆ What is Sticky?

- Switch **automatically learns MAC**
- Saves it in configuration

Enable Sticky

```
switchport port-security  
switchport port-security mac-address sticky
```

Check Config

```
show running-config interface g0/1
```

Important Points

- ✓ Sticky MAC = saved in running config
- ✓ Does NOT age out
- ✓ Must save config to keep permanently

```
copy running-config startup-config
```

Behavior

- Enable sticky → converts dynamic MAC to sticky
 - Disable sticky → converts back to dynamic
-

Real-Life Example

Sticky MAC = **Assigned parking spot** 🚗

- Once assigned → stays fixed
- No one else can use it

MAC Address Table

Types of MAC Entries:

Type	Description
Static	Manual / Sticky
Dynamic	Learned automatically

Show Secure MAC

show mac address-table secure

Command Summary (VERY IMPORTANT)

Show Commands

show mac address-table secure
show port-security
show port-security interface g0/1
show errdisable recovery

Configuration Commands

```
switchport port-security
switchport port-security mac-address <mac>
switchport port-security mac-address sticky
switchport port-security violation {shutdown | restrict | protect}
switchport port-security aging time <minutes>
switchport port-security aging type {absolute | inactivity}
switchport port-security aging static
```

Recovery Commands

```
errdisable recovery cause psecure-violation
errdisable recovery interval <seconds>
```

Final Summary

- ✓ Port Security controls **which devices can connect** 
 - ✓ Protects against **MAC flooding & DHCP attacks** 
 - ✓ 3 modes: Shutdown, Restrict, Protect
 - ✓ Sticky MAC = automatic + saved MAC
 - ✓ Aging helps manage MAC entries
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Quick Revision

- Default MAC allowed = **1**
 - Default mode = **Shutdown**
 - Sticky MAC = **auto save**
 - Restrict = **log + drop**
 - Protect = **silent drop**
 - Shutdown = **port disable**
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Port Security Summary

Core Concept

 **Port Security** is a feature that controls **which MAC addresses are allowed on a switch port**

✓ Only authorized devices allowed

✗ Unauthorized device → action taken 

Why Port Security is Used

 Prevents:

- Unauthorized access
- MAC flooding attacks 
- DHCP starvation attack 

Default Behavior

- Default MAC allowed: **1**
- Default violation mode: **Shutdown**

 Unknown MAC → Port goes **ERR-DISABLED** ✗

Violation Modes (VERY IMPORTANT)

Mode	Behavior
 Shutdown	Port down + logs
 Restrict	Drop + logs + counter
 Protect	Drop silently

Sticky MAC

- ✓ Automatically learns MAC
 - ✓ Saves in running config
 - ✓ Does NOT age out
-

Aging

- ✓ Absolute → time-based
 - ✓ Inactivity → idle-based
-

Basic Configuration

```
interface g0/1
switchport mode access
switchport port-security
```

Verification Commands

```
show port-security
show port-security interface g0/1
show mac address-table secure
```

Conclusion

 Port Security helps:

- Control device access 
- Protect LAN from attacks 
- Limit MAC addresses per port

 Best Practice:

- ✓ Use **Sticky MAC + Restrict mode**
-

Q & A

? Q1: What is Port Security?

✓ Answer:

Port Security is a Layer 2 security feature that allows only specific MAC addresses to access a switch port. If an unknown device tries to connect, the switch takes action based on configured violation mode.

? Q2: Why do we use Port Security?

✓ Answer:

We use Port Security to:

- Prevent unauthorized access 
 - Stop MAC flooding attacks 
 - Protect DHCP server from starvation
-

? Q3: What is default behavior of Port Security?

✓ Answer:

- Allows only 1 MAC
 - Violation mode = shutdown
 - Unknown MAC → port becomes err-disabled
-

? Q4: What is err-disabled state?

✓ Answer:

Err-disabled is a state where the port is shut down automatically due to security violation.

? Q5: Explain violation modes

✓ Answer:

● Shutdown

- Port goes down
- Logs generated
- Manual recovery needed

● Restrict

- Traffic dropped
- Port stays UP
- Logs + counter increase

● Protect

- Traffic dropped
- No logs
- No counter

? Q6: Which mode is best?

✓ Answer:

👉 Restrict mode (because it gives logs + keeps port active)

? Q7: What is Sticky MAC?

✓ Answer:

Sticky MAC automatically learns MAC address and stores it in running configuration.

? Q8: Advantage of Sticky MAC?

✓ Answer:

- No manual MAC entry needed
 - Easy configuration
 - Saves learned MAC automatically
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? Q9: Does sticky MAC age out?

✗ No — it does NOT age out

? Q10: What is MAC aging?

✓ Answer:

Process of removing MAC address after a certain time.

? Q11: Types of aging?

✓ Answer:

Type	Meaning
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Absolute	Removed after fixed time
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Inactivity	Removed if idle
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? Q12: Default aging time?

👉 0 minutes (no aging)

? Q13: Enable port security

```
interface g0/1
switchport mode access
switchport port-security
```

? Q14: Configure sticky MAC

```
switchport port-security mac-address sticky
```

? Q15: Configure restrict mode

```
switchport port-security violation restrict
```

? Q16: Set max MAC addresses

```
switchport port-security maximum 2
```

? Q17: Set aging time

```
switchport port-security aging time 30
```

? Q18: Check port security status

```
show port-security interface g0/1
```

? Q19: Check secure MAC table

```
show mac address-table secure
```

TROUBLESHOOTING QUESTIONS

? Q20: Port in err-disabled — how to fix?

✓ Answer:

Manual:

shutdown

no shutdown

Auto:

errdisable recovery cause psecure-violation

? Q21: Why port security command rejected?

✓ Answer:

Because port is in **dynamic mode**

✓ Must be access or trunk

? Q22: Why port is not coming up?

✓ Answer:

Possible reasons:

- No MAC learned
 - VLAN mismatch
 - Port security violation
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🔥 SCENARIO-BASED QUESTIONS

❓ Q23: What happens if attacker connects device?

✅ Answer:

- MAC mismatch detected
 - Action based on violation mode
 - Port may shut down
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❓ Q24: DHCP starvation attack explanation?

✅ Answer:

Attacker sends multiple fake MAC requests → DHCP IP pool exhausted → legitimate users cannot get IP.

👉 Port Security limits MAC → stops attack

❓ Q25: Can port security stop MAC spoofing?

❌ No fully

✓ Because MAC can be changed

❓ Q26: Difference between static and dynamic MAC?

✅ Answer:

Type	Description
Static	Manually configured
Dynamic	Learned automatically

? Q27: Difference between sticky and static?

✓ Answer:

- Static → manually configured
 - Sticky → automatically learned + saved
-

? Q28: Can we use port security on trunk?

✓ Yes (but mostly used on access ports)

? Q29: Steps to configure port security?

- 1 Set access mode
 - 2 Enable port security
 - 3 Configure MAC/sticky
 - 4 Set violation mode
 - 5 Verify
-

? Q30: Best practice configuration?

switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict

🚀 One Liner

“Explain Port Security simply”

100 Answer:

👉 “Port Security allows only specific MAC addresses to access a switch port and blocks unauthorized devices using different violation modes.”
